```
SELECT tenant_id, idempotency_key, COUNT(*) AS occurrences
FROM payments
WHERE idempotency_key IS NOT NULL
GROUP BY tenant_id, idempotency_key
HAVING COUNT(*) > 1;
```

Explain: This detects existing duplicates; it does not prevent concurrent insertion. Prevention requires an appropriate unique constraint and transaction handling. Null uniqueness behavior differs by database, so define whether keys are required.

Q5. Prevent a lost update

Assume `account_balances(tenant_id, account_id, balance, version)`:

```
UPDATE account_balances
SET balance = balance - :amount, version = version + 1
WHERE tenant_id = :tenant
  AND account_id = :account
  AND version = :expected_version
  AND balance >= :amount;
```

Explain: Require a positive amount, check the affected-row count, and place related ledger writes in the same appropriate transaction. Zero updated rows can mean stale version, insufficient balance, or missing account; resolve deliberately. This statement alone is not a complete payment or transfer implementation.

11. Project stories you should prepare

Use the same five-part structure: **problem -> your responsibility -> implementation -> tradeoff/failure handling -> evidence and learning**. Spend most of the time on what you personally did. Say "we" for team outcomes and "I" for your contribution.

Story A: Java/Vert.x user-management APIs and Redis

Professional opening to adapt: "One relevant project was our commercial user-management platform. My role included [your exact backend responsibility]. The main challenge was [specific problem]. We used Java and Vert.x for the services, and I contributed to [specific implementation]. For frequently accessed data, we introduced Redis caching and measured [verified metric]. The main tradeoff was [freshness, complexity, or another actual concern], which we handled through [actual approach]."

Prepare the details: Trace one endpoint from authentication to response; identify the source of truth; name the client libraries and data model you actually used; explain worker versus event-loop behavior; define cache keys, invalidation and TTL policy; show how a permission change becomes effective; explain tests and the performance measurement.

Questions to rehearse: Why Vert.x? What exactly was cached? Why not cache every response? What happens when Redis is down? What was the baseline for 35%? What did you code yourself? How did you test tenant isolation?

Story B: IAM/MFA and message-signing certificate automation

Professional opening to adapt: "In the reconciliation platform, I worked on [actual security scope]. A recurring operational risk was [specific issue]. I implemented or led [accurate responsibility], with validation and monitoring around [actual steps]. The result was [verified outcome]. My main learning was that automation is complete only when failures are visible and recovery is reliable."

Prepare the details: Identify the identity provider and integration protocol you really used; explain role and resource checks; map certificate inventory to renewal and deployment; explain secure key handling, overlap, verification and rollback. Keep authentication work and certificate work distinct if they were separate projects.

Questions to rehearse: What happens if renewal fails? How do you know the new certificate is being used? What did the 90% effort reduction measure? How are privileges revoked? How are recovery and MFA bypass attempts handled?

Story C: Payment integration or production improvement

Professional opening to adapt: "On a commercial payments project, we needed to [actual business requirement]. I was responsible for [specific technical scope]. We changed [old flow] to [new flow], while preserving [key invariant]. I addressed [failure case] through [actual mechanism]. We validated the change with [tests and rollout evidence] and achieved [verified result]."

Prepare the details: Explain file versus real-time tradeoffs, input validation, duplicate detection, partial failures, reconciliation, monitoring, rollout and operational ownership. If you select an incident instead, explain symptom, evidence, mitigation, root cause, fix and prevention.

Questions to rehearse: What if the request times out after processing? How did you prevent duplicates? How did you reconcile old and new flows? What made rollback safe? Was the $500K recurring savings or avoided cost?

Evidence sheet: fill this in for each story

Field	Your notes
Project and dates	
Business problem and affected users	
My exact responsibility and code ownership	
Request/data flow in 5-7 steps	
Technologies and versions I can verify	
Hardest technical decision and alternative	
One failure scenario and recovery	
Tests and rollout checks	
Metric: baseline, result, period and measurement	
What I would improve now	

If you cannot verify a metric, say: "I don't recall the exact baseline, so I don't want to overstate the percentage. The improvement we observed was [accurate description]."

12. Professional English for the interview

Your goal is clarity, not a particular accent or complicated vocabulary. Speak slightly slower than normal, use short sentences, and pause between the definition, example and tradeoff.

A 60-second introduction

"I'm Sreedhar, a senior engineer with extensive experience in enterprise payments, reconciliation, and client-servicing platforms. At American Express, my work has included Java-based services, API integration, identity and access management, and operational automation. I've also led modernization and reliability improvements across teams. For this role, I bring both implementation experience and an understanding of how production systems behave under real business constraints. I'm particularly interested in contributing through backend engineering, code quality, and practical technical problem-solving."

Edit this to match your actual current responsibilities and motivation. Be ready to give one recent example of code you wrote or reviewed in depth immediately after the introduction.

A reusable technical-answer structure

"[Concept] means [simple definition]. It is useful when [problem]. In a service, I would [mechanism]. The main tradeoff is [cost or limitation]. I would verify it by [test or measurement]."

Example: "Idempotency means repeating a request does not repeat its business effect. It is useful when a client retries after a timeout. I would use a tenant-scoped key and durable uniqueness checks. The main challenge is handling concurrent retries and uncertain downstream outcomes. I would test duplicate submissions and failures at each step."

Useful phrases during coding

- "Let me confirm the input constraints and expected output."
- "Can the input contain duplicates, nulls, or negative values?"
- "I'll start with a correct approach, then discuss the optimization."
- "The invariant is that the current window contains no duplicates."
- "This uses O(n) additional memory to reduce the lookup time."
- "Let me test an empty input and a repeated-value case."
- "I found an issue in the boundary condition. I'll correct it and trace the example again."

Useful phrases during technical discussion

- "The key distinction is visibility versus atomicity."
- "That depends on the consistency requirement and the failure mode."
- "In our implementation, my specific responsibility was..."
- "The alternative was simpler, but it did not meet..."
- "I have not used that feature directly. My understanding is..."
- "I would verify the version-specific behavior before relying on it."
- "Could you clarify whether you mean concurrent threads within one instance or requests across multiple instances?"
- "Let me take a moment to organize the answer."

Avoid: "We used microservices for scalability" with no mechanism; "everything is thread-safe"; "there are no failures"; "exactly once" without a defined boundary; or a metric you cannot explain. Replace broad claims with one concrete implementation and one limitation.

Speaking practice method

Record an answer once. Listen only for three things: Did I answer the question in the first sentence? Did I give a concrete example? Did I stop after the important tradeoff? Repeat the answer in fewer words. Do not try to remove your accent.

13. Timed mock interviews and scoring

Two coding sessions for hours 16-17

Session 1, 60 min: 5 min clarify an unfamiliar variation of sliding window or intervals; 30 min solve; 10 min tests/complexity; 15 min review and rewrite the faulty part. Example variation: find the longest substring containing at most K distinct characters.

Session 2, 60 min: 5 min clarify payment aggregation; 25 min implement; 10 min SQL latest-per-account query; 10 min race-condition or transaction debugging; 10 min review. Include mixed currencies and duplicate records in the discussion.

14. Final recall sheet

Review this during the last 30 minutes of study.

- **Coding:** clarify -> example -> approach -> invariant -> code -> edge tests -> complexity.
- **Design:** KISS = understandable solution; YAGNI = no speculative features; DRY = one source for each business rule.
- **Hash collections:** equals/hashCode contract; stable keys; expected lookup cost; no guaranteed HashMap order.
- **Concurrency:** visibility is not atomicity; compound operations need a consistency boundary; bound queues and dependencies.
- **Java:** immutable versus final; pass-by-value; BigDecimal; comparator correctness; exception propagation.
- **Async:** compose dependent work; do not block event loops; timeout does not necessarily cancel work.
- **Spring:** constructor injection; singleton services should avoid mutable request state; proxy boundaries and rollback rules.
- **Database:** constraints protect invariants; inspect query plans; handle lost updates; keep transactions short.
- **SQL:** tenant scope; currency grouping; NULL semantics; LEFT JOIN filters; deterministic ranking.
- **Cache:** freshness policy; invalidation; stampedes; bounded fallback; cache is not automatically authoritative.
- **Security:** authenticate, then authorize the tenant, resource and action; validate token purpose and claims.
- **Payments:** durable idempotency; safe retries; uncertain outcomes need reconciliation.
- **Production:** customer impact -> evidence -> mitigation -> verify -> root cause -> prevention.
- **Stories:** my contribution, actual mechanism, tradeoff, failure behavior, verified metric.
- **Communication:** answer first, explain simply, give an example, state a limitation, stop.

Prepare two questions for the interviewer: "What are the main backend engineering challenges for this team?" and "How does the team validate reliability and correctness before releasing changes?"

Check the meeting time and time zone, coding environment, audio and connectivity. Keep a blank sheet for examples. Do not start a new topic immediately before the interview.

15. Reference notes

The technical links beside the relevant concepts point to official documentation consulted on October 22, 2026. They are for targeted clarification, not required additional reading. The guide focuses on durable concepts; use your team's actual Java, Spring, Vert.x and database versions for API details and defaults.

Resume inputs reviewed: Sree.pdf (4 pages), Sree-Resume.pdf (3 pages), sree_resume-2.pdf (3 pages), and sree_resume-1.pdf (3 pages), from the archive directory you supplied. Resume outcomes are treated as claims to help you prepare, not independently verified facts. The guide intentionally omits UI preparation and reserves full system-design interviews for the next stage.